



GLOBAL ECONOMICS
GRADE: 11TH
CATCH-UP
CRITERION C6
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SECOND
TERM
2025–2026

Learning objective: Plan sample size to achieve a target margin of error in confidence interval planning.

Assessment criteria:

C6: Establishes the sample size so the error will not exceed a specified amount.

Criteria assessment

Each assessment criterion is evaluated across the problems in this catch-up exam. A criterion is considered passed when it is correctly activated in 11 of the 12 problems of this activity.

Problem 1

Weekly Spending Sample Size

A digital-payment study wants to estimate mean weekly spending by first-job workers. Long-run records show $\sigma = 96$ USD. The team wants 95% confidence with target margin of error $E = 12$ USD. Determine the minimum required sample size.

Problem 2

Utilities Spending Sample Size

A youth housing survey estimates mean monthly utilities spending. The long-run standard deviation is known as $\sigma = 54$ USD. The planner requires 90% confidence and margin of error $E = 6$ USD. Determine the minimum required sample size.

Problem 3

Designer Earnings Sample Planning

A freelance-income team wants the mean weekly platform earnings for young designers. Long-run variability is $\sigma = 125$ USD. The team wants a 95% confidence estimate with margin of error $E = 20$ USD. Extra context says there are 9,800 active designers and last week's median earning was 410 USD. Determine the minimum required sample size.

Problem 4

Ride-Share Interval Width Planning

A transportation budget survey estimates mean monthly ride-share spending by university graduates. The known long-run spread is $\sigma = 72$ USD. The required confidence interval width is $W = 18$ USD at 95% confidence. Determine the minimum required sample size.

Problem 5

Confidence Level Margin Tradeoff

An e-commerce savings study tracks mean monthly discount use by young adults. Long-run standard deviation is $\sigma = 84$ USD. The team wants the same target margin $E = 10$ USD under two confidence levels: 90% and 95%. Compute both required sample sizes and compare.

Problem 6

Delivery Cost Precision Upgrade

A food-delivery cost survey already has $n_{\text{current}} = 140$ users and currently reports margin of error $E_{\text{current}} = 11$ USD at 95% confidence. The organization now wants margin of error $E_{\text{new}} = 8$ USD at the same confidence level. Determine the required total sample size and additional observations needed.

Problem 7

Subscription Margin Reduction Plan

A streaming-subscription survey has $n_{\text{current}} = 121$ and current margin of error $E_{\text{current}} = 9$ USD at 95% confidence. Management requests a 20% reduction in margin of error. Find the new target

margin, required total sample size, and additional observations.

Problem 8

Mobile Banking Precision Stages

A mobile-banking usage survey currently has $n_{\text{current}} = 160$ and margin of error $E_{\text{current}} = 14$ USD at 90% confidence. The policy team wants two staged improvements: first a 10% reduction, then a 30% reduction from the current margin. Determine both required total sample sizes and additional observations.

Problem 9

Nonessential Spending Sample Upgrade

A personal-finance app study has current sample size $n_{\text{current}} = 100$ users. Long-run records indicate $\sigma = 75$ USD for monthly nonessential spending. The confidence level stays at 95%. The team wants to reduce the current margin of error by 20%. First recover the current margin of error, then find the new required total sample size and additional observations.

Problem 10

Side-Income Precision Scenarios

A job-market survey currently has $n_{\text{current}} = 144$ respondents, and long-run salary-spread data give $\sigma = 120$ USD for weekly side-income. Confidence level is 90%. The analysts are considering two reductions from the current margin of error: 25% and 40%. Recover the current margin first, then compute both required sample sizes and compare.

Problem 11

Dual Study Precision Comparison

Two independent youth-economy studies use 95% confidence and want the same 30% margin-of-error reduction. Study A: $n_{A,\text{current}} = 81$, $\sigma_A = 90$ USD. Study B: $n_{B,\text{current}} = 196$, $\sigma_B = 140$ USD. For each study, recover the current margin first, then find the required total sample size and additional observations. Compare which study needs fewer additional observations.

Problem 12

Startup Cost Margin Targets

An entrepreneurship-finance study currently has $n_{\text{current}} = 225$ records, with known long-run standard deviation $\sigma = 180$ USD for monthly startup operating costs. Confidence level is 95%. The team wants to evaluate three reductions from the current margin of error: 15%, 30%, and 50%. Recover the current margin first, then compute all new target margins, required total sample sizes, and additional observations.